ENGR 225: Mechanics 1

Homework Coversheet

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ASSIGNMENT: Design Project, Part 1

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

☐ Minor

■ Major

☐ Extensive

Mark the box that applies (e.g., replace ☐ with ■, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

ENGR 225 Design Project, Part 1: Roadway Selection

Sean Balbale | Assigned: 26 Oct 2026 | Due: 6 Nov 2026

1 Objective

The objective of Part 1 is to select and quantify the roadway that the truss bridge will carry. The specific goals are:

- To classify the roadway and select lane, shoulder, and barrier widths from AASHTO standards.
- To specify the deck section (slab, waterproofing, wearing surface) and calculate material quantities for the 196.9 ft (60 m) span.
- To estimate the embodied carbon of those materials and compare it against lower-carbon alternatives.
- To estimate the raw material cost.

2 Design Basis

Given Parameters

- **Span:** $L = 60\,\text{m} = 196.9\,\text{ft}$, two lanes per direction, no pedestrian traffic.
- **Reference roadway:** Route 2 outside Hartford, a divided freeway with full access control.
- **Classification:** urban principal arterial freeway.
- **Units:** U.S. customary throughout. Emissions are in metric tons (t) of CO_2e, the standard unit for greenhouse gas reporting.
- **Standards:** AASHTO freeway criteria for widths [1]; AASHTO LRFD for the deck slab [2]; the CTDOT Pavement Design Handbook for the wearing surface [3].

3 Roadway Design

Cross-Section AASHTO sets a 12 ft minimum lane width and a 10 ft minimum shoulder, except the left shoulder on a 4-lane freeway, which can be 4 ft [1]. I used the minimums to keep the deck as narrow as possible. Each edge has a 32 in parapet and the median has a 42 in barrier; the barrier base widths (21 in and 24 in) are assumed.

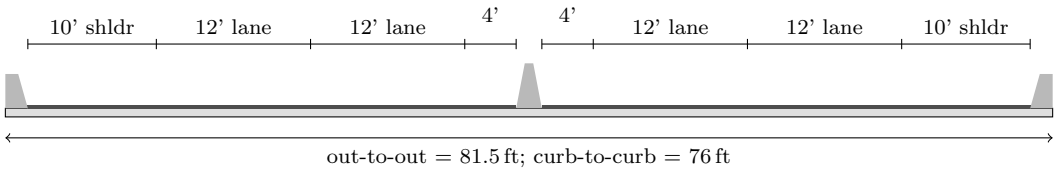


Figure 1: Roadway cross-section: 3 in HMA on membrane over an 8 in reinforced concrete slab (vertical not to scale).

Deck Section

- **Slab:** 8 in cast-in-place reinforced concrete, 2.5 in top cover. AASHTO LRFD requires at least 7 in, and many states use 8 in [2].
- **Waterproofing:** membrane under the HMA [3].
- **Wearing surface:** 3 in hot-mix asphalt (HMA), 1 in of S0.25 over the membrane and 2 in of S0.5 on top. CTDOT recommends S0.25 as the first lift over bridge membranes and caps S0.5 at 2.5 in on bridges [3].

Quantities With out-to-out width $W = 81.5\,\text{ft}$, curb-to-curb width $w_c = 76\,\text{ft}$, parapet area $A_p = 3.00\,\text{ft}^2$, median barrier area $A_m = 4.67\,\text{ft}^2$, and an assumed HMA density of $147\,\text{lb/ft}^3$:

$$V_{\text{deck}} = LWt_s = (196.9\,\text{ft})(81.5\,\text{ft})(0.667\,\text{ft}) = 10{,}696\,\text{ft}^3 = 396\,\text{yd}^3$$
$$V_{\text{barrier}} = L(2A_p + A_m) = (196.9\,\text{ft})(2 \times 3.00\,\text{ft}^2 + 4.67\,\text{ft}^2) = 2{,}100\,\text{ft}^3 = 78\,\text{yd}^3$$
$$W_{\text{HMA}} = Lw_c t_o \gamma_{\text{HMA}} = (196.9\,\text{ft})(76\,\text{ft})(0.25\,\text{ft})(147\,\text{lb/ft}^3) = 548{,}700\,\text{lb} = 274\,\text{tons}$$

Total concrete is $474\,\text{yd}^3$. Tons are U.S. short tons (2,000 lb).

Reinforcement: The bar layout is from the FHWA 8 in deck example [2], with nominal bar weights of 0.668 lb/ft (#4) and 1.043 lb/ft (#5):

Top transverse, #5 @ 8 in:	$(1.043)(12/8) = 1.56\,\text{lb/ft}^2$
Bottom transverse, #5 @ 7 in:	$(1.043)(12/7) = 1.79\,\text{lb/ft}^2$
Top longitudinal, #4 @ 12 in:	$(0.668)(12/12) = 0.67\,\text{lb/ft}^2$
Bottom longitudinal, #5 @ 10 in:	$(1.043)(12/10) = 1.25\,\text{lb/ft}^2$
Total:	$q_{\text{rebar}} = 5.27\,\text{lb/ft}^2$

$$W_{\text{rebar}} = q_{\text{rebar}} LW = (5.27\,\text{lb/ft}^2)(196.9\,\text{ft})(81.5\,\text{ft}) = 84{,}580\,\text{lb} = 42.3\,\text{tons}$$

This leaves out laps, overhang bars, and barrier steel, so it is a lower bound. The FHWA deck sits on girders at 9 ft-8 in, so the real bar sizes depend on how this deck is supported on the truss.

4 Environmental Impact

Embodied Carbon All carbon factors are industry averages. Rebar is 854 kg CO_2e per metric ton (775 kg/ton) from the CRSI EPD, which covers steelmaking, transport to the fabricator, and fabrication (stages A1–A3) [4]. Concrete is $261\,\text{kg/yd}^3$ from the NRMCA Eastern benchmark for 4,000 psi [5], and HMA is 64.5 kg/ton from FHWA's sample EPD [6]. Those two do not list their stages, so I assumed they cover the same ones.

Table 1: Baseline embodied carbon

Material	Quantity	GWP Factor	CO_2e (t)
Concrete (deck + barriers)	$474\,\text{yd}^3$	$261\,\text{kg/yd}^3$	124
Reinforcing steel	42.3 tons	775 kg/ton	32.8
HMA overlay	274 tons	64.5 kg/ton	17.7
Total			**174**

Concrete is 71% of the total, and most of that is cement. The benchmark mix has 475 lb/yd^3 of it [5], and at about 0.9 tons of CO_2 per ton of cement [7], cement is about 74% of the mix's GWP. I left out the membrane since I could not find an EPD for it; at a few millimeters thick it is negligible next to the slab.

Alternatives **1. Slag cement:** The benchmark mix has 82 lb/yd^3 of slag out of 604 lb/yd^3 of cementitious material [5]. Raising slag to 50% removes 220 lb/yd^3 of cement:

$$\Delta\text{GWP} = (220\,\text{lb/yd}^3)(0.9) = 198\,\text{lb/yd}^3 = 89.8\,\text{kg/yd}^3 \quad (-34\%)$$

This ignores the emissions from making the slag, so it slightly overstates the savings.

2. Concrete wearing surface: Swapping the 139 yd^3 of HMA for concrete adds 18.5 t, so HMA is the better choice on carbon too.

Table 2: Embodied carbon of alternatives (same geometry)

Option	**CO_2e (t)**	**Change**
Baseline	174	–
50% slag cement concrete	132	−24%
Concrete wearing surface instead of HMA	193	+11%

Other Impacts The concrete and HMA use several hundred tons of quarried aggregate. None of these figures include trucking to the site, so a lower-carbon supplier far away could end up worse than a closer one. Recycled asphalt pavement (RAP) in the HMA cuts both new aggregate and binder.

5 Raw Material Cost

Costs are for raw ingredients only, not batching, delivery, or labor.

Unit Prices

- **Portland cement:** \$425/ton, and **liquid asphalt binder:** \$625/ton (MassDOT, late 2025) [8]. Slag and fly ash are priced as cement (conservative).
- **Aggregate:** \$18.50 per metric ton (\$16.78/ton) for crushed stone [9], applied to all aggregate.
- **Rebar:** \$920–\$960/ton FOB mill for #5 bar (May 2026) [10]; I used the \$940/ton midpoint.

Calculations The benchmark mix has 604 lb of cementitious material and 2,979 lb of aggregate per cubic yard [5]:

$$\frac{604\,\text{lb}}{2{,}000\,\text{lb/ton}}(\$425) + (1.49\,\text{tons})(\$16.78) = \$128.4 + \$25.0 = \$153/\text{yd}^3$$

For HMA I assumed 5.5% binder by mass, typical for a dense-graded mix:

$$(0.055)(\$625) + (0.945)(\$16.78) = \$34.4 + \$15.9 = \$50/\text{ton}$$

Table 3: Raw material cost (2026 USD)

Material	Quantity	Unit Cost	Cost
Concrete ingredients	474 yd^3	\$153/yd^3	\$72,670
Reinforcing steel	42.3 tons	\$940/ton	\$39,750
HMA ingredients	274 tons	\$50/ton	\$13,780
Total			**\$126,200**

Installed cost will be several times higher once batching, hauling, forming, and placement are included.

6 Conclusion

The roadway is a 4-lane urban freeway deck, 196.9 ft (60 m) long and 81.5 ft wide out-to-out.

1. **Geometry:** four 12 ft lanes, 10 ft right and 4 ft left shoulders, 76 ft curb-to-curb. Widening the left shoulders to 10 ft would add about 15% to the deck and overlay quantities.
2. **Deck section:** 8 in reinforced concrete slab, membrane, and 3 in HMA, giving 474 yd^3 of concrete, 42.3 tons of rebar, and 274 tons of HMA.
3. **Embodied carbon:** 174 t CO_2e, 71% from concrete. A 50% slag mix brings it to about 132 t (−24%).
4. **Cost:** about \$126,200 in raw materials, 58% of it concrete.

Rebar is the number most likely to change, since it comes from a girder-supported deck and the deck supports on the truss are not set yet.

AI Use I used Claude (Anthropic) to locate the AASHTO, CTDOT, EPD, and pricing sources, check the quantity, carbon, and cost arithmetic, and to help edit this report.

References

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[3] Connecticut Department of Transportation, *CTDOT Pavement Design Handbook*, Rep. CT-2307-F-20-1, Jul. 2020. [Online]. Available: https://rosap.ntl.bts.gov/view/dot/57390

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[10] "ISA reports US rebar market recovery accelerating in Q2 2026," *SteelOrbis*, May 1, 2026. [Online]. Available: https://www.steelorbis.com/steel-news/latest-news/isa-reports-us-rebar-market-recovery-accelerating-in-q2-2026-1451268.htm